

ESP32 Firmware Upload Guide

Solar Soil IoT

Instructions for uploading firmware to Sensor Node and Gateway Node boards

Overview

Two ESP32 boards need firmware:

Board	File	Role
Sensor Node	lora/node.ino	Reads sensors, sends via LoRa, deep sleeps
Gateway Node	lora/gateway.ino	Receives LoRa, forwards to MQTT over WiFi

1. Install Required Libraries

In Arduino IDE: **Sketch** → **Include Library** → **Manage Libraries**

Library	Author	Used By
LoRa	Sandeep Mistry	Both
ArduinoJson	Benoit Blanchon	node.ino
DHT sensor library	Adafruit	node.ino
Adafruit INA219	Adafruit	node.ino
PubSubClient	Nick O'Leary	gateway.ino
WiFi	Built-in (ESP32 core)	gateway.ino

ESP32 Board Support:

File → Preferences → Additional Boards Manager URLs:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

Tools → Board → Boards Manager → search "ESP32" → Install

2. Wiring — Sensor Node (node.ino)

ESP32 Pin	Connected To
-----------	--------------

GPIO 4	DHT22 Data
GPIO 36 (ADC0)	Soil Moisture Sensor (analog out)
GPIO 21 (SDA)	INA219 SDA
GPIO 22 (SCL)	INA219 SCL
GPIO 5 (SCK)	LoRa Module SCK
GPIO 19 (MISO)	LoRa Module MISO
GPIO 27 (MOSI)	LoRa Module MOSI
GPIO 18 (SS)	LoRa Module NSS
GPIO 14 (RST)	LoRa Module RST
GPIO 26 (DIO0)	LoRa Module DIO0

3. Wiring — Gateway Node (gateway.ino)

Same LoRa pins as node above. No extra sensors — WiFi handles internet.

4. Configure Before Uploading

node.ino

Line 38 — Change frequency if needed:

```
if (!LoRa.begin(868E6)) // 868 MHz for Europe/India
if (!LoRa.begin(915E6)) // 915 MHz for US/Americas
```

gateway.ino

Line 5-6 — Set your WiFi credentials:

```
const char* ssid = "YourWiFiName";
const char* password = "YourWiFiPassword";
```

Line 18 — Match LoRa frequency with node.

5. Upload Steps

1. Sensor Node: Open `lora/node.ino` in Arduino IDE

- 2. Tools → Board → ESP32 Arduino → select your ESP32 model
- 3. Tools → Port → select the COM port
- 4. Click **Upload** (→ button)
- 5. Open Serial Monitor (Tools → Serial Monitor, 115200 baud) to verify
- 6. **Gateway Node:** Repeat with `loragateway.ino`

6. Verification

Node Serial Monitor output:

```
Sending packet: {"temp":28,"humidity":65,"soil":42,"v":5.2,"current":410}
Going to sleep...
```

Wakes every 15 minutes to send again.

Gateway Serial Monitor output:

```
Connected to MQTT Broker
Forwarding to MQTT: {"temp":28,"humidity":65,"soil":42,"v":5.2,"current":410}
```

7. Troubleshooting

Problem	Fix
LoRa init failed	Check wiring, ensure frequency matches between node & gateway
DHT sensor failed	Check GPIO 4 wiring, try DHT11 instead of DHT22
WiFi not connecting	Verify SSID/password, ensure 2.4 GHz band
MQTT not connecting	Gateway needs internet access via WiFi
No data on dashboard	Run <code>lorasimulator.py</code> to test without hardware